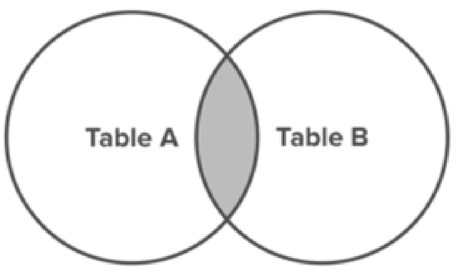
IT1223 (P) - Database Management Systems (P)
Department of Physical Science
University of Vavuniya

Joins In MySQL

Join facilitates the retrieval of information from multiple tables In MySQL several joins' types available.

1. INNER JOIN
2. LEFT OUTER JOIN / LEFT JOIN
3. RIGHT OUTER JOIN / RIGHT JOIN
4. CROSS JOIN
5. SELF JOIN

1. INNER JOIN in MySQL

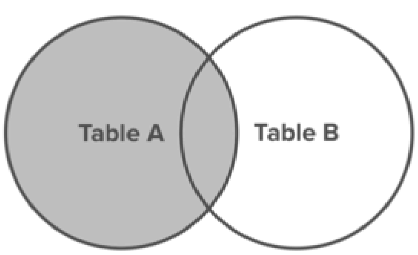


SQL INNER JOIN

- Inner Join returns only the matching rows in both the tables (i e returns only those rows for which the join condition satisfies)
- Syntax:

```
SELECT columns FROM Table_A
INNER JOIN Table_B
ON Table_A.column = Table_B.column;
```

2. LEFT OUTER JOIN / LEFT JOIN in MySQL

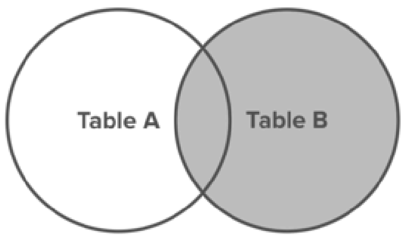


SQL LEFT JOIN

- Left Outer Join Left Join returns all the rows from the LEFT table and the corresponding matching rows from the right table.
- If right table doesn't have the matching record, then for such records right table column will have NULL value in the result.
- Syntax:

```
SELECT column_name(s)
FROM Table_A
LEFT JOIN Table_B
ON Table_A.column_name = Table_B.column_name;
```

3. RIGHT OUTER JOIN / RIGHT JOIN



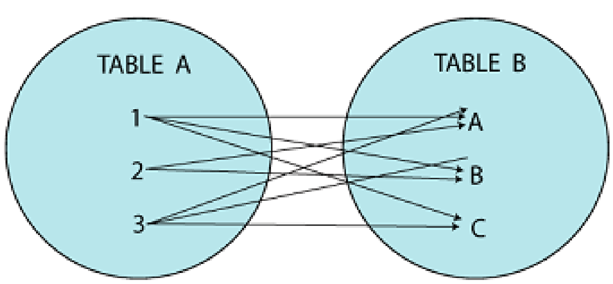
SQL RIGHT JOIN

- Right Outer Join/Right Join returns all the rows from the RIGHT table and the corresponding matching rows from the left table.
- If left table doesn't have the matching record, then for such records left table column will have NULL value in the result.
- Syntax:

```
SELECT column_name(s)
FROM Table_A
RIGHT JOIN Table_B
ON Table_A.column_name = Table_B.column_name;
```

4. CROSS JOIN

CROSS JOIN



- Cross join is also referred to as Cartesian Product For every row in the LEFT Table of the CROSS JOIN all the rows from the RIGHT table are returned and vice versa i.e result will have the Cartesian product of the rows from join tables.
- No of Rows in the Result of Cross Join == (No of Rows in LEFT Table) * (No of Rows in RIGHT Table)
- Syntax:

```
SELECT column_name(s)
FROM Table_A
CROSS JOIN Table_B;
```

5. SELF JOIN

- If a Table is joined to itself using one of the join types explained above, then such a type of join is called SELF JOIN
- Syntax:

```
SELECT column_name(s)
FROM table1 T1, table1 T2
WHERE condition;

OR

Select column_name(s)
FROM student AS S1
INNER JOIN student AS S2;
```